Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

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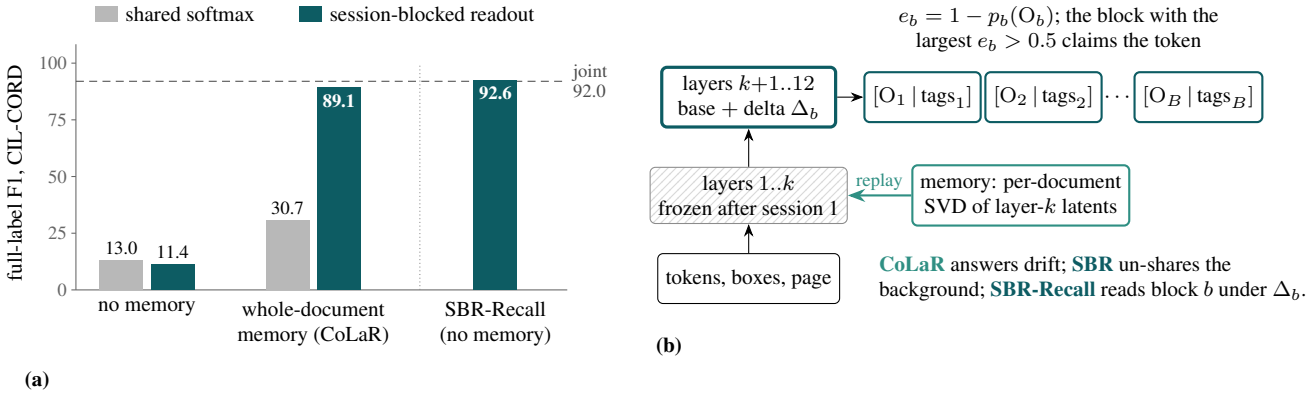


Figure 1. **Two causes of forgetting, and a remedy matched to each.** (a) Class-incremental CORD, LayoutLMv3, full-label F1 (mean of three clean seeds; AA and per-seed values in Tab. 2 and App. D). Left: the 2×2 of Tab. 2; a whole-document memory alone and a session-blocked readout alone stay far below joint training (dashed), together they close most of the gap. Right: SBR-Recall (Sec. 5.3) replaces the memory with one low-rank trunk delta per session. (b) CoLaR replays per-document SVDs of layer-k activations against drift; SBR gives each session its own background logit and routes tokens without task identity; SBR-Recall reads block b under delta Δ_b.

Abstract

Plain replay of a few hundred stored documents reaches 87.9 *average F1 when the input domain shifts under a fixed label space, against* 88.7 *for joint training, but only* 24.3 *when the label space grows, against* 91.3*. We show that continual key-information extraction fails for two separable reasons.* Drift *of the output-facing computation is the familiar one, and a memory repairs it.* Background conflict *arises only under label growth: a stored document labels the entities of later sessions "O", a current document labels those of earlier sessions "O", and one shared background class is pulled both ways, an effect that no memory we tried removes. A factorial on class-incremental CORD separates the two causes: a whole-document memory alone and a session-blocked readout alone both stay far below joint training, together they close most of the gap, and a memory refreshed with the true labels of every class seen so far stays below joint training and below the blocked readout, which never sees those labels (one seed). We give each cause a remedy.* ***CoLaR*** *replays a per-document truncated SVD of mid-network activations and matches raw replay under domain shift without storing raw pixels or token ids.* ***SBR*** *gives each session its own background logit and routes every token without task identity; with CoLaR it reaches* 89.1 *F1 on the full label set, against* 92.0 *for joint training. On a single-schema stream, where label growth itself supplies the routing key,* ***SBR-Recall*** *replaces the memory with per-session low-rank trunk updates and reaches* 92.6*, on par with joint training.*

1. Introduction

Organisations process a continuous stream of visually-rich documents (forms, invoices, receipts) whose meaning depends on the words and on where they sit on the page. Key-information extraction (KIE) is cast as token-level sequence labelling over text, 2-D layout and the page image, and multimodal encoders such as LayoutLMv3 [15], LiLT [41] and

BROS [13] define the state of the art. Deployed systems are not static: new document types appear and schemas grow, while retraining on all history is blocked by cost and by privacy constraints on earlier data. This is continual learning (CL), whose central obstacle is catastrophic forgetting [27].

Replay is empirically the strongest remedy in the CL toolbox [26, 42], and on documents it looks sufficient. On LayoutLMv3, a stream of forms and receipts under one fixed label space is learned by plain replay (ER) with 200 stored documents to 87.9 average F1, against 88.7 for joint training (Tab. 1). Let the label space grow instead, by adding CORD's 30 receipt fields six at a time, and the same method (200 documents) reaches 24.3 session-restricted AA, against 91.3 (Sec. 3). Both ER cells are legacy-protocol runs (†) that early-stop on the test split, which favours ER. Distillation, prompts, low-rank adapters and the continual-NER methods built for the shift of the "O" label do no better (Tab. 3). The usual reading is that class-incremental learning is a harder case of the same forgetting, which a better memory would fix.

We show it is a different failure. Continual KIE fails for two *separable* reasons. **Drift**: the output-facing computation (upper layers and classifier) moves when it is trained on the next task, and in our experiments only old-task supervision of the readout brought it back; this is the known cause, and a memory addresses it. **Background conflict**: under label growth a token labelled "O" means "none of the classes annotated in this session", so stored documents label new entities as background while current documents label old entities as background, and both push the same logit in opposite directions on the same kinds of strings. No memory we tried removes it: refreshing the stored labels with the true labels of every class seen so far, which no deployable method can do, recovers much of the gap but stays below joint training and below the blocked readout introduced here (one seed). The shift of the background is known from class-incremental segmentation [4, 5, 7] and continual NER [30, 50], where it is corrected within one shared background class or, in SSUL, kept away from frozen per-class classifiers. We separate it from drift and give each session its own background, with the trunk and the old blocks kept trainable and re-grounded from a memory.

Figure 1a is the paper in one picture. On class-incremental CORD a whole-document memory with a shared softmax reaches 30.7 full-label F1 and a session-blocked readout without memory 11.4, both far closer to naive fine-tuning (13.0) than to joint training; the two together reach 89.1 ± 2.0, against 92.0 ± 0.4 for joint training. Each cause needs its own remedy, which is what two separable causes predict. When label growth is the only change, the memory itself can go: one low-rank trunk delta per session, read under a session-blocked readout, reaches 92.6 ± 1.0, because each session's background logit tells the model which delta a token belongs to.

Contributions.

1. **Two separable causes of forgetting.** Drift is shared by four encoders, relocated rather than removed by freezing, and in our experiments repaired only by old-task supervision of the readout; background conflict is specific to label growth. A strict 2×2 over memory and readout separates them, and a memory with true labels shows that better memory labels alone do not close the gap (Sec. 4).
2. **SBR and SBR-Recall.** The Session-Blocked Readout gives each session its own background logit, normalised with that session's tags only, and routes every token with no task identity; with a whole-document memory it closes most of the gap to joint training on class-incremental CORD. SBR-Recall stores a low-rank trunk delta per session instead of documents and needs no memory on a single-schema stream (Secs. 5.2 and 5.3).
3. **CoLaR** (Compressed Latent Replay). A document memory must keep whole documents, and each can be compressed on its own: a per-document truncated SVD of mid-network activations matches raw-exemplar replay under domain shift at 60 MB (Sec. 5.1).

2. Related Work

Continual-learning families. Following Wang et al. [42], remedies fall into regularisation (EWC [17], SI [48], MAS [1], LwF [21]), replay (GEM [24], ER [36], DER++ [3], iCaRL [35]), optimisation geometry (GPM [37], OGD [8], C-Flat [2]), methods over a frozen backbone (L2P [45], DualPrompt [44], CODA-Prompt [38], O-LoRA [43], CL-LoRA [12], RanPAC [28]) and architectural isolation [25, 47]. Replay is empirically the strongest [26, 42] but stores raw data. Exemplar-free alternatives replay class-conditional feature statistics (PASS, FeTrIL, FeCAM [9, 32, 56]) or merge models and realign the classifier (LCA [39]).

Background shift. In class-incremental semantic segmentation a pixel labelled background at one step may belong to an old or a future class. MiB [4] revises the cross-entropy, matching a background label with the probability of background or any old class, and the distillation, matching the old model's background with background or any new class; PLOP [7] pseudo-labels the background with the old model's confident predictions and distils pooled multi-scale features; SSUL [5] adds an unknown class inside the background, also pseudo-labels the background, freezes the backbone and the past classifiers, keeps a tiny exemplar memory, and replaces the softmax with per-class sigmoids, because under noisy background targets a softmax's relative scores let the background out-score frozen old classes.

In continual NER the same shift of "O" is handled by distilling the old model's soft targets on "O" tokens (ExtendNER [30]), by pooled feature distillation with confidence-based pseudo-labels (CPFD [50]), by causal-effect distillation (CFNER [54]) or by entity-triplet generation [46]. These methods correct the supervision of one background class per model. SSUL is the closest to ours: its frozen per-class sigmoids already keep old classes out of training on stale background labels. SBR differs in three ways: it keeps the upper trunk and the old blocks trainable and re-grounds them from a memory with labels that are valid for each block; it groups the classes by session, with one softmax and one background logit per block; and it routes each token by the blocks' background claims. Unlike PLOP, CPFD and SSUL it needs no old-model pseudo-labels.

Task identity, isolation, and latent replay. Kim et al. [16] decompose class-incremental learning into within-task prediction and task-identity prediction; adapter methods that add one module per task avoid test-time task identity by combining the modules' features (EASE [55]) or folding them into the pre-trained weights (InfLoRA [22]). SBR predicts task identity explicitly, per token, and on documents its detector is trained with real negatives at no extra cost, since every document of a session contains tokens that are not that session's entities. In continual NER, ICE [23] trains per-session classifiers on a frozen encoder and SpanKL [52] uses per-type span heads with distillation; neither replays stored examples. REMIND [11] stores product-quantised mid-level CNN feature maps per image and Latent Replay [31] each example's activations at a chosen layer, both replayed through the plastic upper layers; generative replay [40] samples hidden representations from a class-conditional generator. Their examples carry one label each; a document carries one label per token, bound to a position, which CoLaR keeps by compressing each document with a truncated SVD.

Where forgetting lives; continual document IE. Ramasesh et al. [34] localise forgetting in image classifiers to the layers nearest the output, and Zhai et al. [49] find that multimodal LLMs forget under fine-tuning even with a frozen image encoder. Continual relation extraction uses episodic memory with prototypes [6, 10, 53]; like continual NER it is sentence-level and text-only. Document-level continual learning is nascent: GDI-Bench [20] limits forgetting in multimodal-LLM fine-tuning by freezing most parameters, chosen by update magnitude; continual table detection [29] uses plain ER; and ProtoNER [19], the closest work to ours, adds key-information classes to a LayoutLM-family model from a few labelled documents with prototypical networks. None of these document-level works separates the causes of forgetting.

3. Setup

Task and scenarios. A document is a triple of OCR tokens, normalised boxes and the page image; the model emits BIO tags per token and is scored by entity-level F1 (`seqeval`). We use FUNSD (149/50 train/test forms), CORD (800/100/100 receipts, 30 classes) and SROIE (626/347 receipts). **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema; only the input distribution shifts. **CIL-CORD**: CORD's 30 classes arrive six at a time over five sessions; the head grows, and tokens of every class outside the session, earlier or later, are labelled "O". **Mixed**: six sessions over three datasets (two class-incremental FUNSD sessions, a shift to SROIE, two class-incremental sessions on CORD's super-classes, a FUNSD revisit with all its labels), so boundaries alternate new classes and new domains. App. A gives the sessions and pool sizes.

Backbones and methods. LayoutLMv3-base [15] is the primary encoder; LiLT [41], BROS [13] and text-only BERT are secondary. Heads are linear. The comparison set is fixed in advance: naive fine-tuning and joint training as references, and one published method per family: ER [36] and DER++ [3], LwF [21], CODA-Prompt [38], CL-LoRA [12], LCA [39], and the continual-NER methods CPFD [50] and IS3 [33]. Replay baselines keep 200 raw documents (DER++ $\alpha{=}\beta{=}0.5$); in clean-protocol cells ER instead keeps as many raw documents as our memory. LwF uses $T{=}2$. Our CL-LoRA port trains a single rank-8 adapter on the attention projections of the frozen encoder, shared by all sessions, unlike the published one-adapter-per-task design, so its numbers describe this port. Our memory keeps 50 documents per session (per task on DIL).

Training protocol. AdamW, learning rate $5{\times}10^{-5}$, weight decay 0.01, gradient clipping 1.0, batch size 2, three seeds (42, 7, 123). Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100) with early stopping, patience 2, on a 10% split held out from that session's *training* data; the test split is never used for model selection. Cells marked † in the tables come from earlier runs (the *legacy protocol*), which early-stop on the session's test split (except joint training on Mixed) and vary in epoch cap (10 to 100) and precision (fp16, fp32, or mixed across seeds). Every cell of the $2{\times}2$ and the joint-training reference on CIL-CORD are clean. A cell switches to the clean protocol only when all three clean seeds exist and never combines the two; App. G lists the settings of every † cell.

Scoring. On CIL-CORD the headline number is the *full-label F1*: the final model on the complete CORD test set with all 30 classes, plain `seqeval`, the usual evaluation of

continual NER. Only clean-protocol runs record it, so tables give it for the runs completed so far, with the number of seeds marked; a comparison with a cell that has only AA (every † cell, DIL, Mixed) is made in AA on both sides and labelled so. Retention needs per-session test sets, from whose retention matrix $R[i, j]$ (F1 on session j after training session i) we report average accuracy $\text{AA} = \frac{1}{T}\sum_j R[T, j]$ and backward transfer $\text{BWT} = \frac{1}{T-1}\sum_{j<T}(R[T, j] - R[j, j])$. A session's test labels show only that session's classes, and scoring raw predictions against them is a pitfall: a model that *correctly* labels another session's entity is charged a false positive in every session that does not own it, so a jointly trained model reads 33. We score each session *session-restricted*: a predicted tag outside session j's label set becomes "O" before span matching, without re-taking the argmax, so an error is charged at most once and a confusion between sessions still costs a miss. On every seed both metrics order the same groups of clean runs: the two without a memory, then CoLaR with a shared softmax, then the true-label refresh (seed 42, one seed so far), then CoLaR-SBR, SBR-Recall and joint training; runs inside a group can swap places. Restricted AA is the more lenient metric for the runs without a memory (by 4.8 to 12.3 points) and for CoLaR (1.2 to 5.2), and within 1.3 of full-label F1 for CoLaR-SBR, SBR-Recall and joint training (App. B, which also works one receipt through every scoring).

4. Two Causes of Forgetting

4.1. Cause 1: drift, and why a memory is needed

Drift is the known cause and we treat it briefly; App. F holds the full evidence. Under naive fine-tuning on FUNSD→CORD→SROIE the first task collapses after one further task (FUNSD 89.1 → 3.3 on LayoutLMv3, 83.9 → 2.1 on LiLT, 88.9 → 6.6 on BROS). Per-token CKA [18] between consecutive checkpoints falls with depth, from 1.00 at the embeddings to 0.25 at the last layer and 0.16 at the classifier, with the same profile on all four backbones (Fig. 2, left), as Ramasesh et al. [34] found for image classifiers; old-task-weighted parameter displacement is largest at the head but not in the late encoder layers, so "upper layers" is a statement about representations (App. F). *Freezing relocates the drift.* Freezing the head and late layers on DIL gives BWT −76.8, the naive level, while early-layer CKA, ≈1.0 under naive training, falls to 0.19; freezing everything gives BWT 0.0 and learns nothing new (AA 41.1; Fig. 2, right; seed 42). *Only old-task supervision of the readout repaired it.* On average replay brings every backbone close to joint training while EWC and LwF stay near naive training (Tab. 1), and five families we built or ported that keep updating a shared network without stored old-task targets (head merging, merging with classifier realignment [39], subspace transfer, isolated head slots, feature-Gaussian replay)

Table 1. **Drift and its remedy are shared by four encoders** (DIL, AA, mean ± s.d., three seeds; all cells from our first grid, legacy protocol, App. G). On average replay comes within one point of joint training or exceeds it, and EWC and LwF stay within ten points of naive training. "–": did not fit the 6 GB GPU.

Backbone	Naive	EWC	LwF	ER	DER++	Joint
LayoutLMv3	41.3±1.5	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9	88.7±1.1
LiLT	40.1±0.5	48.4±5.9	–	85.7±0.5	84.4±0.3	83.8±1.2
BROS	40.3±1.8	45.0±4.9	35.1±3.1	87.8±0.4	87.5±1.1	87.5±1.3
BERT	39.4±0.3	46.6±3.0	40.5±0.1	77.1±0.9	77.4±0.6	77.2±0.4

reach at most 66.0, against 41.3 for naive training and 88 for replay (App. F gives each failure reason). On a fixed label space, then, no method we tried that updates a shared trunk and readout without old-task supervision recovers the old tasks; this is why the remedies below keep a memory, and Sec. 5.3 shows where isolation can replace it.

4.2. Cause 2: background conflict under label growth

True memory labels help but do not close the gap. The same whole-document memory that retains under domain shift reaches only 30.7 full-label F1 on CIL-CORD with a shared softmax (Tab. 2). The obvious suspect is the memory's labels: a document stored at session t carries "O" on every entity introduced after t. We therefore refresh the stored labels at every boundary with the documents' true labels for every class seen so far, a control no deployable method can run; a refresh at a boundary cannot cover the session trained next, whose entities stay "O" in the stored documents while it is trained. The refresh rewrites 105/1019/1394/1239 stale tokens at the four boundaries (seed 42) and lifts full-label F1 from 32.4, the same memory without the refresh, to 79.1, and AA from 33.6 to 65.8 (clean protocol, seed 42, one seed so far). That is below joint training on the same seed (91.7 and 91.1) and below CoLaR-SBR, which never sees those labels, on every seed. The two metrics disagree on how much it recovers: 79% of the gap between CoLaR and joint training in full-label F1, 56% in AA. The reason is in the retention matrix: sessions 3 to 5 are learned only to 2 to 15 F1 on their own diagonal, sessions 3 and 4 rise to 75 to 83 one session later, and the last session ends at 14.7. AA weights the five sessions equally, whereas full-label F1 counts entities, and the last session holds 8% of the test entities (App. E).

The conflict on one receipt. Take a receipt stored at the end of session 1, when only menu fields exist: its item prices are labelled `menu.price` and its total price is "O". In session 5 a current receipt labels its total price `total.total_price` and its item prices "O", since menu fields are not annotated in that session. Replayed in the same batch, the two documents supervise the one background logit of a shared softmax on price strings that



Figure 2. **Drift is output-facing, and freezing moves it.** Left: linear CKA between checkpoints before and after the second task falls with depth on all four backbones (1.0 = no drift; naive fine-tuning, three seeds). Right: freezing the head and late layers leaves backward transfer unchanged, (a) for any number of frozen layers and (b) on three DIL task orders and a ten-session CIL-CORD stream (seed 42, no memory).

look alike, the stored one pulling totals towards "O" and the current one pulling item prices towards it. The refresh removes the first pull only for the classes known at the last boundary, and current documents keep the second. Restricting each loss to the classes its labels can speak for shows that both pulls matter (AA): slicing the replay loss to the classes alive at storage gives 42.0 (two fp16 rescore seeds; 39.4 on the fp32 gate run of seed 42), slicing the current loss as well 60.5 on average over three fp32 gate seeds, and model pseudo-labels added on both sides, in the spirit of the pseudo-labelling methods, 44.1 (one seed). No label-side repair we ran exceeds 69.3 AA on any seed, below every seed of CoLaR-SBR; most of these repairs are legacy gate or rescore runs, whose test-split early stopping favours them (App. E).

What is new. That the background changes meaning under label growth is known (Sec. 2); most methods built for it keep one shared background and correct its supervision. In no label-side repair we ran did better labels close the gap: with the true labels in the memory a shared softmax stays below joint training and below the blocked readout (one seed), and the pull of the current documents that remains is the background shift MiB corrects within a shared softmax, which we did not run (Sec. 7). We therefore remove the shared background rather than repair it (Sec. 5.2), and the next experiment shows that this, and not the memory, is what label growth needs.

4.3. The two causes separate

Table 2 crosses a remedy for each cause on the same encoder, stream and recipe. Without a memory, blocking the readout leaves full-label F1 near naive training (13.0 to 11.4; AA 18.3 to 21.6): the blocks' labels are consistent, but the trunk drifts and the old sessions are lost (App. E). With a whole-document memory and a shared softmax drift is contained but the labels conflict (30.7). Only the cell with both closes most of the gap, on every seed: 89.1±2.0 full-label F1 and 88.4±2.6 AA, against 92.0±0.4 and 91.3±0.8 for joint training. Each remedy alone recovers little and the two together recover most of the gap, which is what two separable causes predict; in every configuration we ran, the conflict was resolved only at the readout.

Table 2. **Memory × readout on CIL-CORD** (LayoutLMv3; mean ± s.d. over three seeds, per-seed values when fewer). Rows: how drift is handled; columns: how the background is supervised. Only the cell with both remedies approaches joint training, under either metric. Every cell is clean protocol (Sec. 3).

	Shared softmax		**Session-blocked readout**	
	full-label F1	AA	full-label F1	AA
No memory	13.0±0.2	18.3±0.5	11.4±2.3	21.6±2.2
CoLaR memory	30.7±1.5	33.5±1.0	**89.1±2.0**	**88.4±2.6**
Joint training (reference)	full-label F1 92.0±0.4, AA 91.3±0.8			

5. Remedies

5.1. CoLaR: a memory for drift

A memory repairs drift by re-exercising the readout on inputs that keep their old labelling. To find the least it must store we vary the stored object and hold the mechanism fixed: activations are captured at the output of layer k, layers $1..k$ are frozen after the first task, and stored activations are replayed through the plastic upper layers and head (DIL, LayoutLMv3; single-seed diagnostics unless stated).

Whole documents retain; summaries collapse. Replaying the stored activations of 50 whole documents per task retains AA 87.3. Summaries of the *same* store collapse to the naive level: a rank-16 pooled spectral summary 41.9, per-class Gaussians 39.4, real k-means centroids 36.7–39.7. A single-variable control isolates the ingredient: four whole documents retain 63.8, and the identical token features placed on layout carriers from other documents retain 36.7. The binding of a token's feature to its position and its label is worth 27 points. The requirement is specific to token-level documents: on image class-incremental learning, where an image carries one label, a class-conditional Gaussian head (SLCA [51]) summarises the readout and beats the same latent memory (App. F).

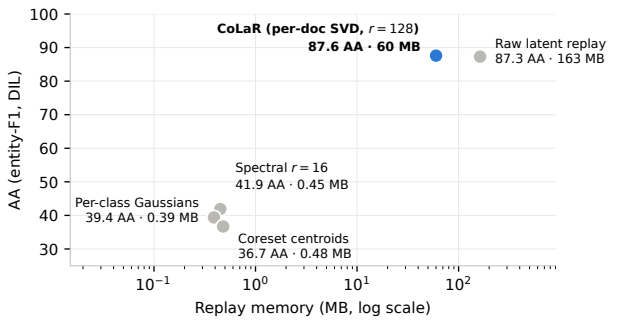


Figure 3. Memory size versus retention (DIL, LayoutLMv3, AA, seed 42; log-scale memory). Per-document SVD (CoLaR) matches raw latent replay at 2.7× less storage; every summary that breaks whole-document binding falls to the naive level.

Per-document compression. The pooled feature space is close to full rank, but one document's tokens share a layout and a vocabulary, so its activation matrix $H_j \in \mathbb{R}^{N_j \times d}$ (N_j tokens, width d) is low-rank. CoLaR (Compressed Latent Replay) stores a rank-r truncated SVD per document, $H_j \approx U_r \Sigma_r V_r^\top$, with the document's boxes, attention mask and labels, rebuilds $\hat{H}_j$ at replay, and injects it at layer k; the loss is cross-entropy on the current batch plus cross-entropy on replayed documents. At k=4, r=128 and 50 documents per task it retains 87.8±0.5 AA on DIL in 60 MB (1 MB = 10^6 bytes throughout), against 87.9±1.4 for ER with 200 raw documents and 88.7±1.1 for joint training, and is 2.7× smaller than the uncompressed latent memory it matches (87.3 at about 163 MB; Fig. 3); r=64 gives 80.6 at about 32 MB. The memory holds no raw pixels or token ids. We make no formal privacy claim: mid-network activations can leak content. Rebalancing the memory does not move AA, and read-side additions lower it (App. F).

5.2. SBR: a readout for the background conflict

Every session that brings new tags opens a block b holding those tags $\mathcal{T}_b$. A shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$. The Session-Blocked Readout (SBR) gives every session its own background logit $o_b(x) = w_b^\top h(x) + c_b$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\big([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\big), \tag{1}$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block b a token labelled "O" in session-b data is truly background, so the block's supervision is never stale. For a token x with label y in a document of session s,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\, \mathrm{CE}\big(p_b(\cdot \mid x),\, y_b\big), \qquad y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: m_{sb}=1 when session s annotates every tag of $\mathcal{T}_b$ (the block is supervised fully); m_{sb}=1 with y_b=O_b when b belongs to a different annotation schema: the document is annotated under its own schema, so no token carries a tag of b, and we treat every token as a negative for b (a training-time convention; related schemas such as SROIE and CORD share entity kinds like totals and dates, so on Mixed it can label a real entity of another schema as background); and m_{sb}=0 otherwise, because the document's "O" labels say nothing about block b. Stored documents follow the rule of the session that stored them. A session that adds no tags creates no block, so on a fixed label space SBR has a single block. In the receipt example of Sec. 4.2 the stored receipt now trains only block 1 and the current one only block 5, so neither pulls the other's background.

Inference needs no task identity. Each block reports how strongly it claims the token, $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest e_b labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise. The trunk is shared across blocks and kept stable by the CoLaR memory. Each step sums the block terms of Eq. (2) over the current batch and four replayed CoLaR documents; after the first session layers $1..k$ are frozen, and at the end of each session 50 of its documents are compressed and stored. App. C gives the pseudo-code (Alg. 1), the supervision mode of every (session, block) pair and the memory accounting.

5.3. SBR-Recall: isolation in place of memory

With SBR the memory is written during training and never read at test time. SBR-Recall replaces it by isolation: it stores, for each session $b > 1$, a low-rank delta Δ_b [14] of the layers above k (rank 64 on the attention projections and feed-forward layers), freezes these layers after the first session (the base θ_0; Δ_1=0), and computes block b under the trunk state that learned it:

$$p_b(\cdot \mid x) = \mathrm{softmax}\big([\, o_b,\, z_{\mathcal{T}_b}\,]\big)\big(x;\, \theta_0 + \Delta_b\big). \tag{3}$$

Every term of Eq. (2) is computed under the delta its block is read under, at training and at test time. Without a memory on a single-schema stream no term of a later session involves an old block, so neither the base, nor its delta, nor its rows of the readout receive a gradient after its session: each old block's e_b is unchanged up to weight decay, and only competition from new blocks can move its tokens (App. C). A session that adds no block reuses the delta of the first block it trains fully. Inference costs one pass through the upper layers per

Table 3. **Comparison on LayoutLMv3** (session-restricted AA on CIL-CORD and Mixed, and full-label F1 on CIL-CORD; mean ± s.d. over three seeds, per-seed values when fewer; "–": not run). †legacy protocol; $^{n=k}$: k clean seeds so far; ^{c}one fp16 LwF seed that collapsed to all-background is excluded; ^{b}with the 50-document memory. Settings per cell: App. G.

Method	CIL-CORD full-label F1	CIL-CORD AA	Mixed AA	DIL AA
Naive	13.0±0.2	18.3±0.5	44.4/43.7†	41.3±1.5†
ER	–	24.3±2.4†	75.6/73.3†	87.9±1.4†
LwF	–	18.4/18.5†,c	47.7†,c	43.7±1.1†
DER++	–	20.4±1.4†	75.1/74.0†	88.0±0.9†
CODA-Prompt	–	14.0/15.9†	12.1/10.6†	32.5±1.2†
CL-LoRA (2025)	–	18.7/18.6†	45.7/43.9†	41.1±0.3†
CPFD (2023)	–	2.7/0.0†	0.3/15.1†	6.1±7.3†
IS3 (2024)	–	5.6/7.7†	0.0/40.6†	38.3†
LCA (2026)	–	13.7/7.0†	38.0/36.4†	43.9±2.5†
CoLaR	30.7±1.5	33.5±1.0	75.6/74.6†	87.8±0.5†
CoLaR-SBR	89.1±2.0	88.4±2.6	**89.8**/**89.6**†	86.2±1.3†
+ LexSlot head slots	–	89.7±0.5†	89.6±1.0†	87.0±0.4†
SBR-Recall	**92.6±1.0**	**91.9±0.6**	85.9†,b	–
Joint training (reference)	92.0±0.4	91.3±0.8	86.9/88.5†	88.7±1.1†

delta; App. C gives the parameterisation and the pseudo-code (Alg. 2).

Why isolation works here, and where it stops. Isolation normally lacks a routing key: a shared-softmax counterpart would have to combine or merge the deltas at test time (as EASE and InfLoRA do with their modules) or predict the session with a separate model; we did not run one, and our CL-LoRA port, one adapter shared by all sessions, is a baseline of Tab. 3, not such a counterpart. Under label growth SBR supplies the key: each session's background logit, trained on that session's true negatives, tells the model which delta a token belongs to. Where there is no such key, isolation does not replace memory. On a fixed label space there is one block and nothing to route by. Across schemas the all-background terms of Eq. (2) have no positive counterweight without stored documents: on Mixed, SBR-Recall without a memory scores 44.7 (seed 42), with other-schema blocks silent until their domain returns, and 85.9 with the 50-document memory restored (App. D).

6. Experiments

Only a blocked readout closes the class-incremental gap. Table 3 reports the fixed comparison set. On CIL-CORD every continual method with a shared softmax stays at or below CoLaR in AA (33.5), and the two blocked readouts close most of the gap, SBR-Recall all or nearly all of it on every seed: CoLaR-SBR 89.1±2.0 and SBR-Recall 92.6±1.0 full-label F1 against 92.0±0.4 (AA 88.4±2.6 and 91.9±0.6 against 91.3±0.8); SBR-Recall is within 2.1 points of joint training on every seed under either metric. Four baselines

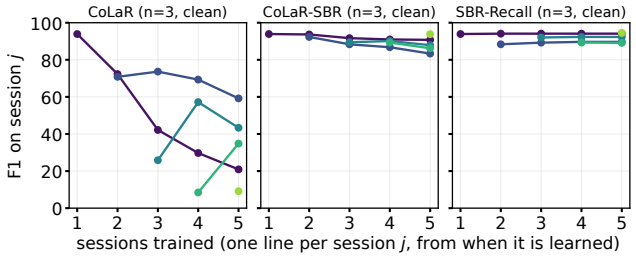


Figure 4. **Retention per session on CIL-CORD** (session-restricted F1 on session j after each later session, mean over the seeds; each panel title gives their number n and the protocol, † = legacy). With a shared softmax the curves fall steeply and late sessions are learned poorly; with the blocked readout most sessions hold; under SBR-Recall no session falls more than 0.6 points below its diagonal on any seed.

end below naive training on every seed, and the retention matrices show why: they learn later sessions poorly. CPFD learns no session after the first (at most 0.0 F1 on a later diagonal, in fp32 as in fp16), and IS3, CODA-Prompt and LCA each learn some later session on every seed to at most 32.3 F1 on its diagonal, where naive training learns every session to at least 78.6; like naive training they keep little of the earlier sessions (App. G). These rows are our ports onto one growing token head (App. G), not checked against their published NER numbers. The table repeats the 2×2 at the scale of the whole comparison: the methods that address only drift (ER, DER++, CoLaR) are strong on DIL and stay far below joint training on CIL-CORD, and neither the methods built for the shift of "O" (CPFD, IS3) nor those that keep a frozen encoder with a shared softmax (CODA-Prompt, our CL-LoRA port) recover CIL-CORD. On Mixed, where both causes act, CoLaR-SBR reaches 89.7 against 87.7 for joint training and 74.5 for ER. On DIL, where only drift is present and SBR has one block, CoLaR, CoLaR-SBR and ER all come close to joint training on average (87.8, 86.2, 87.9 against 88.7); SBR is below CoLaR there, and its only differences are the separate background logit and the thresholded decision. App. D lists the per-seed values behind every cell.

Where the remaining gap lies. Figure 4 shows it per session. With a shared softmax the curves fall steeply and irregularly, some sessions partly recovering at a later boundary, and the later sessions are learned poorly even on their own diagonal. With the blocked readout every session is learned and most keep nearly all of their F1, but the losses are uneven, with one early session losing the most. Under SBR-Recall the curves are flat: no session loses more than 0.6 points on any seed, and the larger moves, up to 2.1, are gains. App. D gives the matrices.

Ablations. *Plasticity and memory are both needed.* The blocked readout without a memory scores 21.6 AA, and with every encoder layer but the last frozen after the first session, and no memory, 14.8 (one seed; sessions after the second are barely learned): on a shared trunk the blocks need both plasticity and a memory that keeps the trunk stable (App. E). *Calibration is not the gap.* Tuning the claim threshold and a per-block bias on the test set itself changes AA by at most 0.6 (seed 42), so what SBR still loses to joint training is retention of early sessions (Fig. 4), the part SBR-Recall removes. *The delta needs capacity, not more isolation.* For SBR-Recall without a memory, rank 8 on the attention projections alone gives 82.5 AA (seed 42) with flat retention but two sessions stuck at 65 and 68 F1; rank 64 with the feed-forward layers learns them on all three seeds, in legacy runs under the protocol of rank 8 and in the clean runs of Tab. 3 (App. D). *Extra head slots are redundant with SBR*: stacking LexSlot on CoLaR-SBR moves the mean by less than two points on every scenario (App. D).

Cost. CoLaR-SBR stores 50 compressed documents per session (100.7 MB on CIL-CORD, App. C) and one background logit per session. SBR-Recall stores no documents on a single-schema stream and the deltas of the four later sessions (57 MB if stored in half precision, 113 MB in fp32 as trained and evaluated), and pays one upper-layer pass per session at inference (about 3.7× the encoder compute of one forward pass at five sessions and k=4; App. C). Per-run training times are in App. H.

7. Limitations

The study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones; SBR and SBR-Recall are evaluated on LayoutLMv3. Cells marked † early-stop on the test split, as stated in Sec. 3. The freezing experiment, the memory-content ladder and the true-label refresh are single-seed diagnostics; the tables are three-seed unless a cell shows fewer values. Joint training is a reference, not a bound: with early stopping it can be matched or slightly exceeded. We did not port MiB or SSUL to token labelling; CPFD, which carries pooled feature distillation and background pseudo-labels to continual NER, is in the comparison set. SBR-Recall grows by one delta and one inference pass per session, freezes its base on the first session, so its accuracy can depend on which session comes first, and does not replace memory on a fixed label space or across annotation schemas; its Mixed result is one seed. SBR's supervision rule uses the annotation schema of each session, which is known at training time and is not used at test time. CoLaR is a replay method whose memory grows with the number of sessions; its settings were chosen on LayoutLMv3 and transfer to BROS but trail raw replay by 5–9 points on average on the text-heavy BERT and LiLT. CoLaR stores no raw pixels or token ids, which is weaker than privacy.

8. Conclusion

Replay retains documents under domain shift and fails under label growth because continual key-information extraction forgets for two separable reasons. Drift is answered by a memory, which can be a compressed whole-document one (CoLaR). Background conflict is not solved by a better memory: with the true labels in the memory a shared softmax stays below joint training and below a blocked readout that never sees them (one seed); it is answered at the readout, by giving each session its own background logit (SBR), and where label growth supplies the routing key that readout lets per-session deltas replace the memory (SBR-Recall). A method for continual document understanding should now be evaluated against both causes, on a stream that grows its label space, with a mask-free metric, and against a memory that already has the true labels.

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